

# Chapter Three

## Crystal Binding

### 3.3 Bulk Modulus and Compressibility

The bulk modulus is given by  $\beta = -V \left( \frac{dP}{dV} \right)$  and compressibility is defined as  $\frac{1}{K} = \beta = -V \left( \frac{dP}{dV} \right)$

From first law of thermodynamics  $dQ = dU + dW$  at absolute zero  $dQ = 0$

So  $\frac{dU}{dV} = -P$ ,  $\frac{d^2U}{dV^2} = -\frac{dP}{dV}$  so  $\frac{1}{k_0} = V_0 \left( \frac{d^2U}{dV^2} \right)$  where  $V_0$  is volume of crystal at  $R = R_0$

For NaCl crystal  $R = R_{cl} + R_{Na} = \frac{a}{2}$  and volume per molecule is  $= \frac{a^3}{4}$

Therefore  $V = \frac{a^3}{4} N = 2NR^3$  so  $\frac{dV}{dR} = 6NR^2 \Rightarrow \frac{dR}{dV} = \frac{1}{6NR^2}$

$$\frac{dU}{dV} = \left( \frac{dU}{dR} \right) \cdot \left( \frac{dR}{dV} \right) \Rightarrow \frac{d}{dV} \left( \frac{dU}{dV} \right) = \left( \frac{dU}{dR} \right) \cdot \frac{d}{dV} \left( \frac{dR}{dV} \right) + \left( \frac{dR}{dV} \right) \cdot \frac{d}{dV} \left( \frac{dU}{dR} \right)$$

$$\left( \frac{d^2U}{dV^2} \right) = \left( \frac{dU}{dR} \right) \cdot \left( \frac{d^2R}{dV^2} \right) + \left( \frac{dR}{dV} \right) \frac{d}{dR} \left( \frac{dU}{dR} \right) \cdot \left( \frac{dR}{dV} \right)$$

$$\left(\frac{d^2U}{dV^2}\right) = \left(\frac{dU}{dR}\right) \cdot \left(\frac{d^2R}{dV^2}\right) + \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right)$$

At equilibrium separation

$$U = N \left[ -\frac{\alpha q^2}{4\pi \epsilon_0 R} + B \exp\left(-\frac{R}{\rho}\right) \right] \text{ For } \left(\frac{dU}{dR}\right) = 0 \text{ and } V = V_0 = 2NR_0^3$$

$$\text{So } \left(\frac{d^2U}{dV^2}\right) = \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right) \text{ so } \frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right)$$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right) \Rightarrow \frac{2R_0^3}{36R_0^4} \left(\frac{d^2U}{dR^2}\right) = \frac{1}{18NR_0} \left(\frac{d^2U}{dR^2}\right) = \frac{\alpha q^2 N}{R_0^3} \left(\frac{R_0}{\rho} - 2\right)$$

$$\frac{1}{K_0} = \beta = \frac{\alpha q^2}{18R_0^4} \left(\frac{R_0}{\rho} - 2\right) \Rightarrow \frac{R_0}{\rho} = \frac{18R_0^4 \beta}{\alpha q^2} + 2.$$

Bond dissociation energy of ionic molecule is given by

$DE = -U_{total} - IP + EA$ . where  $U_{total}$  potential energy,  $IP$  is ionisation potential and  $EA$  is electron Affinity.

**Example:** if  $\frac{B}{R^m}$  is repulsive term like  $\frac{\rho}{R_0}$  then prove that value of  $m = 1 + \frac{72\pi \epsilon_0 R_0^4}{K_0 \alpha q^2}$  and

$$U = -\frac{N\alpha q^2}{4\pi \epsilon_0 R_0} \left(1 - \frac{1}{m}\right)$$

**Solution:**  $U(R) = -N \left[ \frac{\alpha q^2}{4\pi \epsilon_0 R} + \frac{B}{R^m} \right] \quad V = V_0 = 2NR_0^3$

Hence,  $\frac{dU}{dR} = 0$  at  $R = R_0 \Rightarrow R_0^{m-1} = \frac{mB4\pi \epsilon_0}{\alpha q^2} \Rightarrow R_0 = \left(\frac{4\pi \epsilon_0 mB}{\alpha q^2}\right)^{\frac{1}{m-1}}$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right), \quad \frac{dR}{dV} = \frac{1}{6NR^2} \quad \text{Where } \frac{d^2U}{dR^2} = N \left( -\frac{2\alpha q^2}{4\pi \epsilon_0 R_0^3} + \frac{m(m+1)}{R_0^{m+2}} \right)$$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{1}{6NR^2}\right)^2 \cdot N \left( \frac{m(m+1)B}{R_0^{m+2}} - \frac{2\alpha q^2}{4\pi \epsilon_0 R_0^3} \right)$$

$$\frac{1}{K_0} = \frac{1}{18R_0^4} \cdot \left( \frac{m(m+1)B}{R_0^{m-2}} - \frac{2\alpha q^2}{4\pi \epsilon_0} \right) \Rightarrow \frac{1}{K_0} = \frac{1}{18R_0^4} \cdot \left( \frac{m(m+1)B\alpha q^2}{mB4\pi \epsilon_0} - \frac{2\alpha q^2}{4\pi \epsilon_0} \right)$$

$$\Rightarrow \frac{1}{K_0} = \frac{\alpha q^2}{72\pi \epsilon_0 R_0^4} [m+1-2] \Rightarrow m = 1 + \frac{72\pi \epsilon_0 R_0^4}{K_0 \alpha q^2}$$

Total energy,  $U(R) = -N \left[ \frac{\alpha q^2}{4\pi \epsilon_0 R} + \frac{B}{R^m} \right]$  put  $R_0^{m-1} = \frac{mB4\pi \epsilon_0}{\alpha q^2}$  and  $U = \frac{-\alpha q^2}{4\pi \epsilon_0 R_0} \left[ 1 - \frac{1}{m} \right]$